

# NGUYEN KHAC BAO

AI Operations Intern – Chatbot, Automation & Workflow Integration

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## CAREER OBJECTIVE

AI Operations Intern candidate focused on building practical AI chatbots, automation workflows, CRM-style data flows, and lightweight reporting tools. Hands-on with Python, prompt engineering, FastAPI/Streamlit, REST API integration, RAG chatbot workflows, Telegram alerting, Docker, and cloud deployment.

## EDUCATION

**Posts and Telecommunications Institute of Technology (PTIT)**

2022 – 2027

*Bachelor of Information Technology*

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

## TECHNICAL SKILLS

**AI / Chatbot:** Gemini API, OpenAI-compatible APIs, RAG pipelines, prompt engineering, chatbot workflow, text summarization

**Automation / Integration:** Python scripting, REST API integration, Telegram Bot API, event-driven alerts, workflow documentation

**Backend / Tools:** FastAPI, Streamlit, JSON/API integration, WebSocket, Node.js basics, RabbitMQ

**Data / Reporting:** Pandas, CSV/JSON/JSONL processing, SQL basics, MySQL/Cloud SQL, Qdrant Cloud

**DevOps / Cloud:** Docker, Linux, Git/GitHub, Google Cloud Run, AWS S3/SageMaker, Vast.ai, Hugging Face

**AI / CV / OCR:** OpenCV, OCR, YOLOv8n, object detection, tracking, dataset QA

## LANGUAGES

Vietnamese: Native | English: TOEIC 610/990; technical reading proficiency for documentation, API references, model cards, and engineering articles.

## PROJECTS

**Vietnamese Legal RAG Chatbot**

2026

*Personal Project | Python, FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL, Docker, Cloud Run | Source: [GitHub](#)*

- Built and deployed a **RAG chatbot** with **Streamlit** frontend and **FastAPI** backend as separate **Cloud Run** services.
- Indexed 32,980 Vietnamese QA/context records into **Qdrant Cloud** using **Gemini embeddings** for retrieval-grounded answers.
- Implemented **chat APIs**, **Cloud SQL** chat history, and Secret Manager-based credential handling.
- Designed the architecture to support internal documents, SQL-backed records, async indexing, and OpenAI-compatible LLM serving.

**AI Alerting & Telegram Automation Tool**

2026

*Personal Project | Python, Scapy, Random Forest, CNN1D, Telegram Bot API, Logging | Source: [GitHub](#)*

- Built a real-time **event-driven alerting pipeline** with packet capture, windowed feature aggregation, AI inference, and automated notification workflow.
- Implemented live local simulation with HTTP POST burst traffic, Scapy-based capture, runtime Random Forest detection, and **Telegram alert automation**.
- Trained and evaluated CNN1D, BiLSTM, Isolation Forest, and One-Class SVM on CICIDS2017; achieved **0.997 AUC** with CNN1D after threshold tuning.

**Mini Microservices Social Web**

2026

*Personal Project | React, Node.js, Express Gateway, RabbitMQ, PostgreSQL/Cloud SQL, Docker Compose | Source: [GitHub](#)*

- Built a mini social web system with **API Gateway**, user/post/comment/feed/notification services, JWT-protected routes, and **RabbitMQ** event messaging.
- Containerized frontend, gateway, backend services, RabbitMQ, and database configuration using **Docker Compose**.

**AutoFall Labeler Tool**

2026

*Personal Project | Python, Streamlit, OpenCV, YOLO-Pose, Pandas | Source: [GitHub](#)*

- Built a **human-in-the-loop workflow tool** with Streamlit review UI, metadata/annotation management, YOLO-Pose keypoint suggestions, and reviewed-only export.
- Added QA reports for auto-label accuracy, correction rate, pose missing rate, keypoint edit rate, and noisy real-world data cases.

## RELEVANT STRENGTHS

- Built practical AI tools with Streamlit/FastAPI interfaces, API-based LLM integration, and Docker/cloud deployment.
- Comfortable processing CSV/JSON data, designing metadata schemas, generating QA reports, and documenting workflow outcomes.
- Able to prototype lightweight AI modules and automation flows involving chatbot/RAG, notifications, OCR, and human validation.

## ACTIVITIES

**AI Research Group – University Club**

2025 – Present

*Member*

- Discussed AI applications, workflow automation ideas, Machine Learning, Computer Vision, and RAG/LLM systems.